

ADVANCED WEB = EVENING

EXERCISES

1: ABC Company, located in Kigali City, has recently upgraded its web-based Employee Management System developed using PHP and MySQL. As part of the upgrade, the Human Resource (HR) department requires a new feature that allows authorized staff to add new employees into the system database. Each employee record should contain an Employee ID, Employee Name, Position, and Salary.

During system testing, several issues were discovered. Some users submit the form without completing all required fields. Others enter non-numeric values in the Salary field, which causes errors during data processing. In addition, there are cases where users attempt to insert an Employee ID that already exists in the database. This leads to database conflicts and prevents the system from storing the new employee record correctly.

You are required to develop the “Add Employee” feature for the Employee Management System so that staff members can properly insert employee information into the database while ensuring that data is validated and errors are handled appropriately.

Questions

Answer the following questions by writing complete and functional PHP code.

- a. Create an HTML/PHP form that allows the HR administrator to enter Employee ID, Employee Name, Position, and Salary. The form must submit the data using the POST method.

- b. Display a JavaScript alert message indicating whether the employee record was inserted successfully or if an error occurred.
- c. Write PHP code that executes the SELECT query and displays the retrieved records in a styled table.

2: You are developing an Employee Absence Monitoring System for a company. The system records the number of days an employee has been absent from work and assigns a status based on the absence period.

If the employee is absent for 0–2 days, the status should be Normal.
If the employee is absent for 3–5 days, the status should be Warning.
If the employee is absent for 6–10 days, the status should be Manager Review.
If the employee is absent for more than 10 days, the status should be HR Action Required.

QUESTIONS

- A. Write a JavaScript script that prompts the user to enter the number of days the employee was absent.
- B. Use conditional statements to determine the employee's absence status according to the rules provided.

3: You are part of the development team for the Library Fine Management System. The administrator wants a PHP script that calculates fines for overdue books using a function.

TASKS:

a) Create a PHP function named `calculateFine()` that accepts one parameter `$daysLate`. Use conditional statements to return the fine amount based on these rules:

- A. 0–3 days: 0 RWF
- B. 4–7 days: 500 RWF
- C. 8–14 days: 1000 RWF
- D. More than 14 days: 2000 RWF

b) Modify the function so that it also returns a message string combining the fine amount and the number of days late.

Example: php Book is 5 days late. Fine: 500 RWF

End